LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 5

Course Code: CSE276
Course Title: ARTIFICIAL INTELLIGENCE FOUNDATIONS
Course Outcome: CO5
Paper: Sample Subjective Questions
Maximum Marks: 10 x 10 = 100

Unit: 5
Unit Title: Generative AI, Responsible and Explainable AI
Unit Mapped RBT: L2 (Understand)
Total Questions: 10
Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Explain what generative Artificial Intelligence is and describe three of its applications. (10 Marks)

(a) Describe what distinguishes a generative model from a model that classifies or predicts.

(b) Describe three applications, stating for each what the input is and what the output is.

[L2 - Understand]

Q2. Describe the components that make up a well-designed prompt. (10 Marks)

(a) Describe what the role, the task, the context, the constraints and the output format each contribute.

(b) Explain which components would fix a reply that was too long and drifted off the subject.

[L2 - Understand]

Q3. Explain how a Retrieval-Augmented Generation pipeline works, stage by stage. (10 Marks)

(a) Describe what happens to the documents before any question is asked, and what happens when one is.

(b) Explain what has to be done when one of the source documents is amended, and why.

[L2 - Understand]

Q4. Describe hallucination in a large language model and the strategies that reduce it. (10 Marks)

(a) Describe what a hallucination is and explain what about the model's training causes it.

(b) Describe two mitigation strategies, stating the residual risk that each one leaves behind.

[L2 - Understand]

Q5. Explain why a deployed decision system needs to be explainable. (10 Marks)

(a) Describe two distinct reasons an explanation is required, naming who needs it in each case.

(b) Explain what an explanation must let a reader do that a bare output does not.

[L2 - Understand]

Q6. Describe how bias enters a model trained on historical records, and how it can be detected. (10 Marks)

(a) Describe the routes by which bias reaches the model, including the effect of proxy features.

(b) Explain how the bias would be detected before release, and what evidence you would present.

[L2 - Understand]

Q7. Explain the privacy and the security risks of using a public large language model service for work data. (10 Marks)

(a) Describe two privacy risks and two security risks that arise.

(b) Explain the controls an organisation should put in place before permitting such use.

[L2 - Understand]

Q8. Describe what Artificial Intelligence governance requires of an organisation. (10 Marks)

(a) Describe what governance decides before a system is built and before it is released.

(b) Explain what should continue to happen after the system is in service, and why.

[L2 - Understand]

Q9. Explain the ethical and societal implications of generative Artificial Intelligence. (10 Marks)

(a) Describe three implications, drawn from misinformation, employment and consent.

(b) Explain one measure that would reduce the harm in one of the three cases you described.

[L2 - Understand]

Q10. Distinguish prompt engineering from fine-tuning as ways of improving the answers an assistant gives. (10 Marks)

(a) Describe what each of the two changes, and what each one costs to carry out.

(b) Explain which of the two you would try first for a poorly performing assistant, and why.

[L2 - Understand]

----- End of Sample Question Set -----